Fault Logs

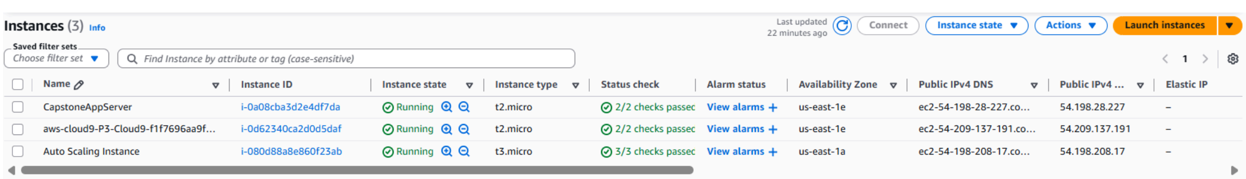
FAULT 1 LOG

Fault Description: Terminate the Auto Scaling Instance behind the ALB. This removes the only instance currently registered and healthy in `CapstoneTG`.

Failure Category: Compute failure

Hypothesis: Upon termination, the ALB will continue routing traffic to the instance for up to 30 seconds until the health check detects it as unhealthy. During this window, requests will receive 502/504 errors, causing a spike in the `HTTPCode_Target_5XX_Count` metric. The `HealthyHostCount` widget will drop from 1 to 0. The `SLO-Breach-5XX-Error-Rate` alarm may fire depending on traffic volume during the failure window. The ASG should automatically detect the terminated instance and launch a replacement within 2–3 minutes. Full recovery, new instance passing health checks and `HealthyHostCount` returning to 1, is expected within 5 minutes of injection. The home page and the `/students` endpoint will be unavailable during the unhealthy window.

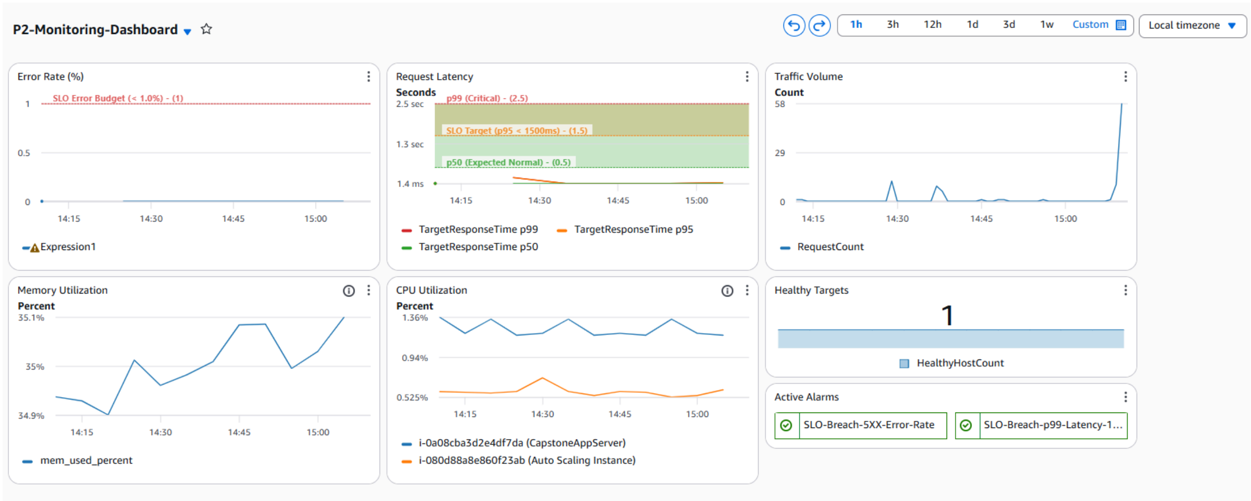
Pre-injection state:


Instances (3) Info

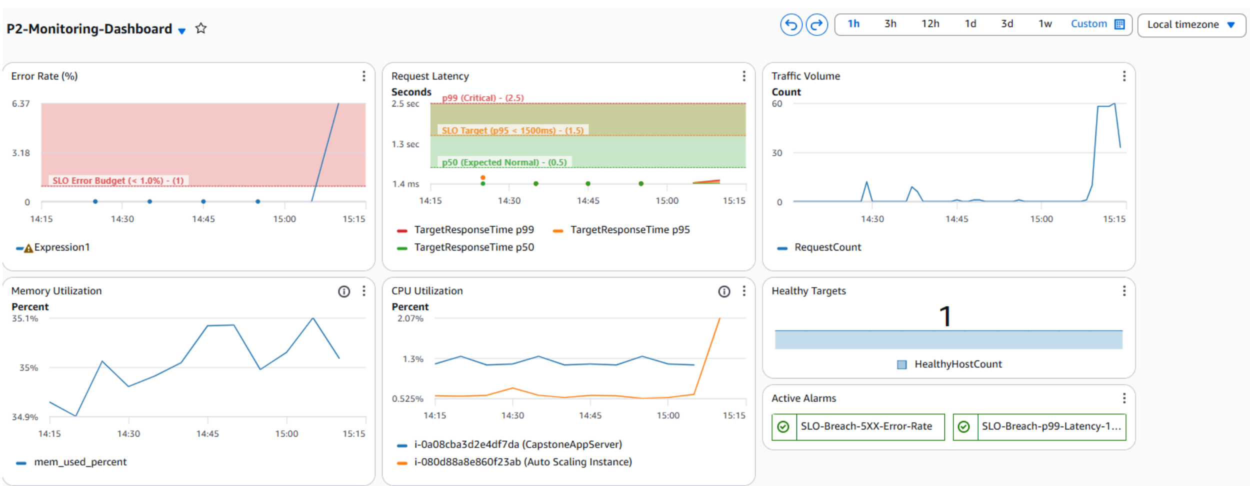
Last updated 22 minutes ago | Connect | Instance state | Actions | Launch instances

Saved filter sets: Choose filter set | Find Instance by attribute or tag (case-sensitive)

Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IPv4 DNS	Public IPv4 ...	Elastic IP
CapstoneAppServer	i-0a08cba3d2e4df7da	Running	t2.micro	2/2 checks passec	View alarms +	us-east-1e	ec2-54-198-28-227.co...	54.198.28.227	–
aws-cloud9-P3-Cloud9-f1f7696aa9f...	i-0d62340ca2d0d5daf	Running	t2.micro	2/2 checks passec	View alarms +	us-east-1e	ec2-54-209-137-191.co...	54.209.137.191	–
Auto Scaling Instance	i-080d88a8e860f23ab	Running	t3.micro	3/3 checks passec	View alarms +	us-east-1a	ec2-54-198-208-17.co...	54.198.208.17	–



Injection Timestamp: 3:14 PM 5/3/2026



This Message Is From an External Sender
This message came from outside your organization.

You are receiving this email because your Amazon CloudWatch Alarm "SLO-Breach-5XX-Error-Rate" in the US East (N. Virginia) region has entered the ALARM state, because "Threshold Crossed: 2 out of the last 3 datapoints [13.565891472868216 (03/05/26 19:15:00), 6.367041198501873 (03/05/26 19:10:00)] were greater than the threshold (1.44) (minimum 2 datapoints for OK -> ALARM transition)." at "Sunday 03 May, 2026 19:27:20 UTC".

View this alarm in the AWS Management Console:
https://urldefense.com/v3/__https://us-east-1.console.aws.amazon.com/cloudwatch/deeplink.js?region=us-east-1*alarmsV2:alarm*SLO-Breach-5XX-Error-Rate__;Iy8!!Jx1HYqWJHNgT5A!n4Rv3pwGYf4nRoqCgVL9kiPnkmZNM-K3vWjfvuTcKo-S3kfi5ibKKPS-0mIe8Xcmu-TTQi2tmM3pjvMFVKjD5Nkcwb6BiQ$

Alarm Details:
- Name: SLO-Breach-5XX-Error-Rate
- Description: WHAT: The web application is returning an HTTP 5XX error rate greater than 1.44%, breaching the availability SLO.

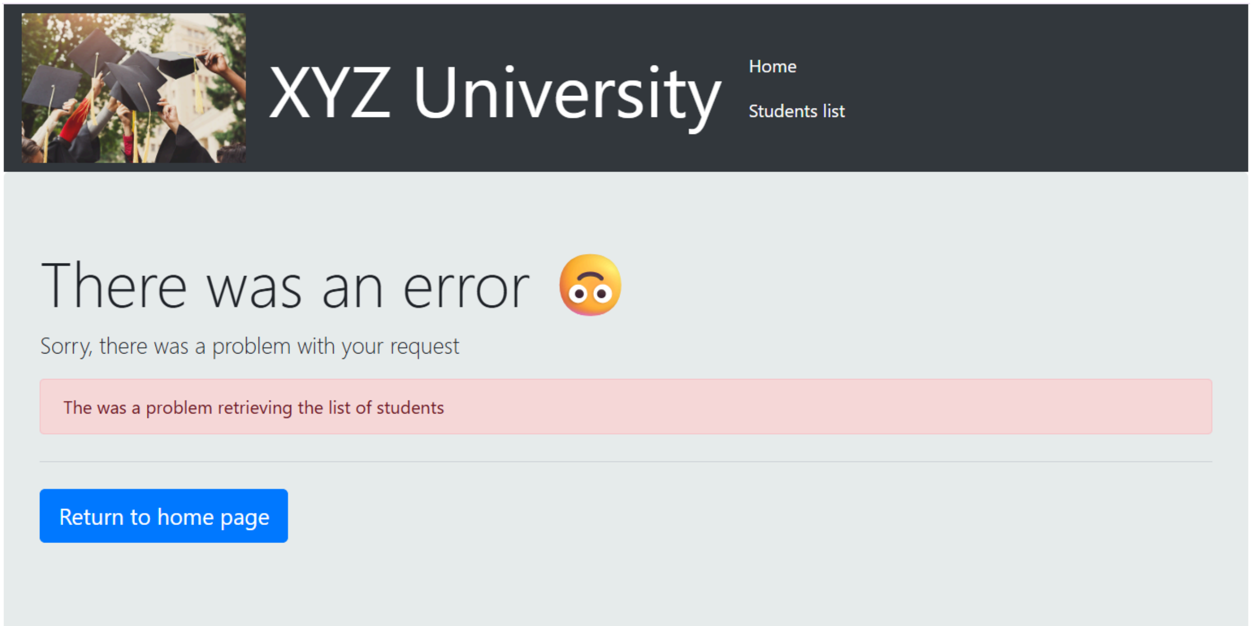
WHY: This indicates a severe backend failure, likely EC2 compute exhaustion or database connection dropping, preventing users from accessing student records.

WHERE: Check ALB Target Group health, EC2 CPU Utilization, and P2-ApplicationLogs for stack traces.
- State Change: OK -> ALARM
- Reason for State Change: Threshold Crossed: 2 out of the last 3 datapoints [13.565891472868216 (03/05/26 19:15:00), 6.367041198501873 (03/05/26 19:10:00)] were greater than the threshold (1.44) (minimum 2 datapoints for OK -> ALARM transition).
- Timestamp: Sunday 03 May, 2026 19:27:20 UTC
- AWS Account: 040458019306
- Alarm Arn: arn:aws:cloudwatch:us-east-1:040458019306:alarm:SLO-Breach-5XX-Error-Rate

Threshold:
- The alarm is in the ALARM state when the metric is GreaterThanThreshold 1.44 for at least 2 of the last 3 period(s) of 300 seconds.

Monitored Metrics:
- MetricExpression: (m1 / m2) * 100
- MetricLabel: Expression1

State Change Actions:
- OK:
- ALARM: [arn:aws:sns:us-east-1:040458019306:SRE-OnCall-Alerts]
- INSUFFICIENT_DATA:



Reversal Plan: No manual action required. ASG minimum capacity is 1 and will auto-replace. If no replacement instance appears in **EC2** → **Instances** within 5 minutes, manually set Desired Capacity to 1 in **EC2** → **Auto Scaling Groups** → `CapstoneAutoScalingGroup` → **Edit**.

FAULT 2 LOG

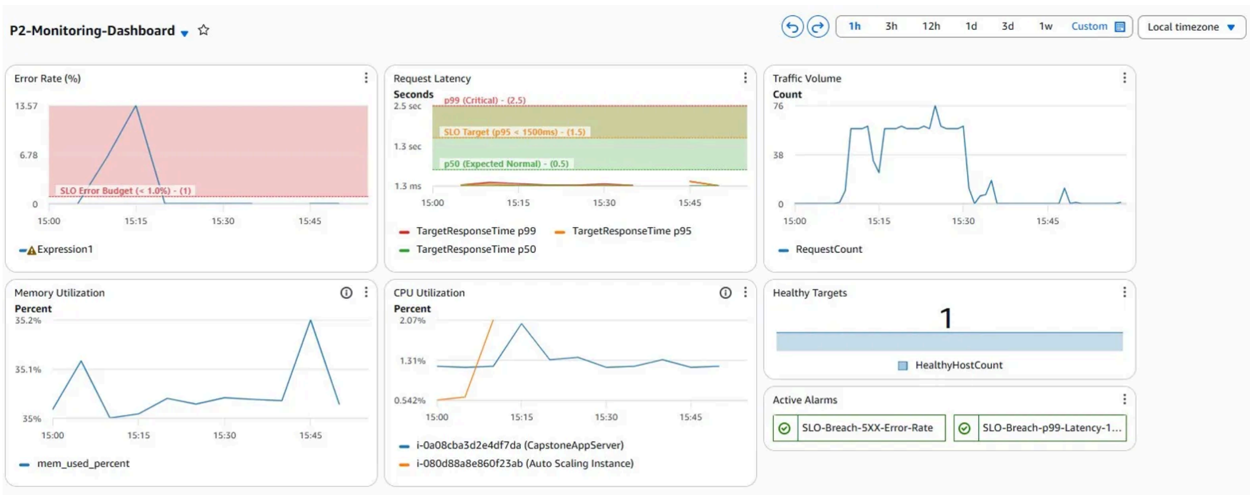
Fault Description: Reboot the RDS instance `CapstoneDB` via the RDS console. This causes a database outage of approximately 5-15 minutes during which the Node.js application cannot execute any database queries.

Failure Category: Database failure

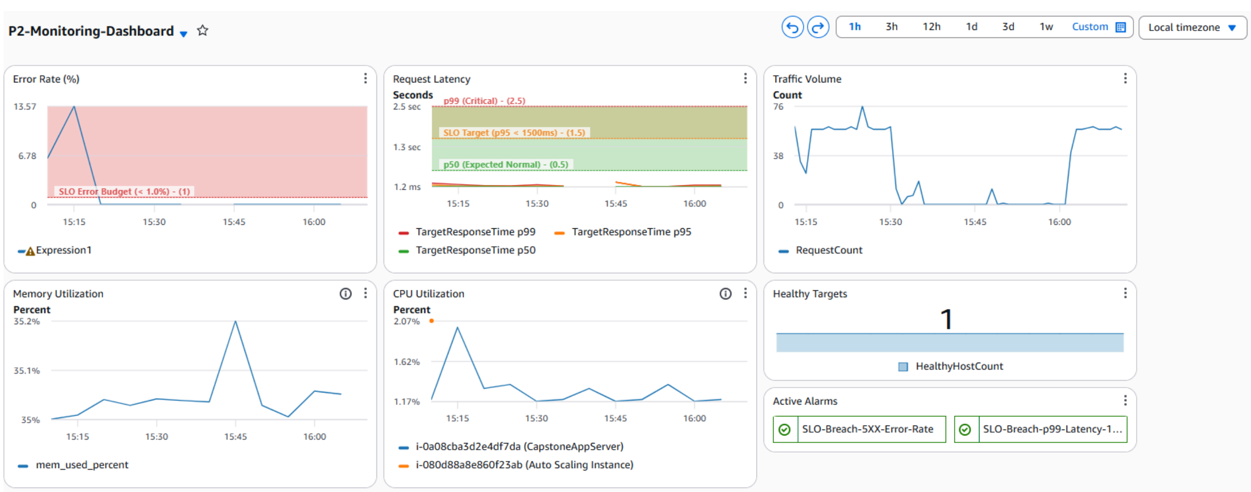
Hypothesis: During the RDS reboot, `GET /` (the home page) will continue returning HTTP 200 because it serves static HTML with no database dependency. However, `GET /students` will fail with HTTP 500 because the Node.js application cannot connect to RDS and crashes the request. This will only generate 5XX metrics at the ALB if active requests are made to `/students` during the outage; passive monitoring alone will not detect this failure. A curl loop against the `/students` endpoint will be run from Cloud9 during the fault to ensure 5XX events are generated. The `SLO-Breach-5XX-Error-Rate` alarm should fire within one evaluation period (approximately 5-10 minutes) once the curl loop is running. The `Request

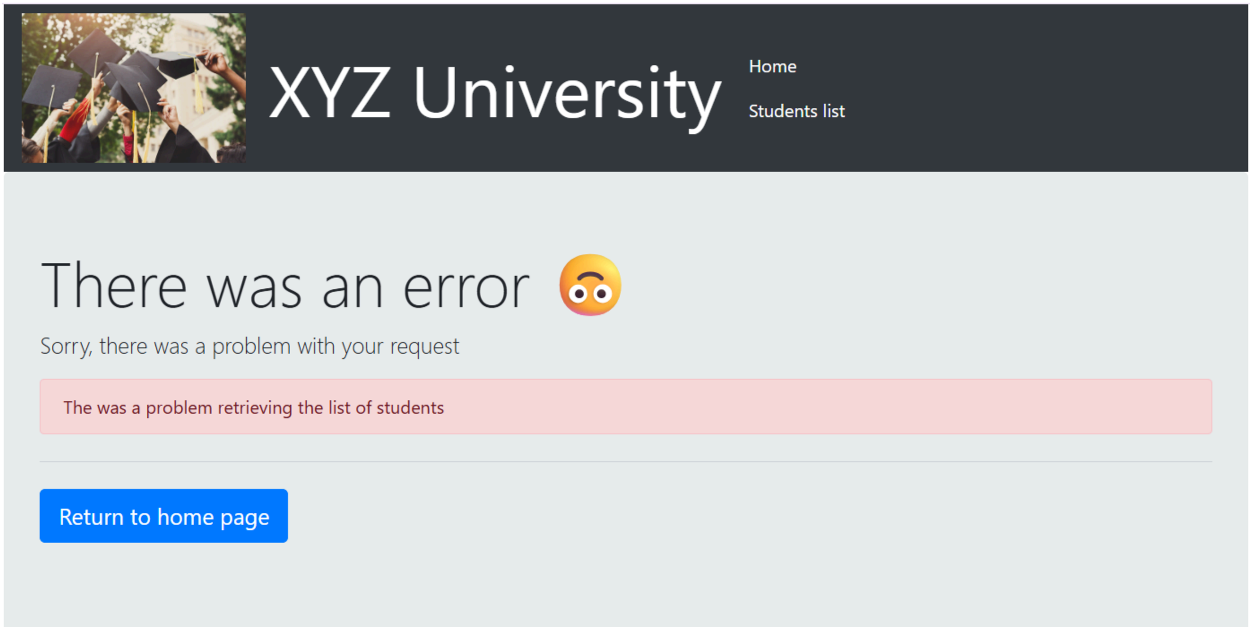
`Latency` widget may also spike briefly as connection timeout errors accumulate before the app returns 500. The dashboard will not show any RDS-specific metric change; this is a known observability gap. Full recovery is expected 5-15 minutes after the reboot completes when RDS returns to Available status.

Pre-injection state:



Injection Timestamp: 4:05 PM 5/3/2026





Reversal Plan: No manual action required. RDS will automatically return to Available. Monitor **RDS** → **Databases** → `CapstoneDB` status. If not Available within 20 minutes, check **RDS** → `CapstoneDB` → **Logs & Events tab** for errors.

Post: Everything back to normal in 3h display.